



OPEN CV 5 · AGENTIC VISION

Evidence before explanation.

A reproducible OpenCV 5 pipeline that turns visual changes into sealed evidence, then lets a bounded agent rerun, seal, complete, or request human approval.

38 / 38

local tests passed

2 TP · 2 TN

seeded smoke evaluation

SHA-256

canonical evidence receipt

STATUS

LOCAL + CONTAINER + BOUNDED AWS RUN VERIFIED



ONE EVIDENCE PATH

From video bytes to bounded action.

Observation, interpretation, and action are separate stages. The agent never substitutes prose for visual evidence.

**CORE INVARIANT**

A later explanation may cite only known frame IDs and must preserve the action selected by the OpenCV evidence policy. Unknown citations or unapproved recommendations fail closed.

SUBSTANTIVE VISION PIPELINE

Decode. Compare. Localize. Seal.

OpenCV 5.0.0.93 is pinned and used across decoding, comparison, localization, and review rendering.

1

Decode

VideoCapture reads frames and frame rate.

2

Preview

Grayscale + INTER_AREA downsampling.

3

Compare

Normalized absdiff produces a change score.

4

Localize

Threshold, morphology-close, contours, rectangles.

5

Stabilize

Hysteresis suppresses continuous-motion floods.

6

Review

Separate overlay; source frames stay unchanged.

ADVERSARIAL FIX

One public 125-frame clip initially produced 56 cards. Hysteresis reduced that to one without disabling re-arming after a quiet frame.



EVIDENCE CHANGES THE ACTION

Four actions. No invented fifth.

The planner is deterministic and fail-closed. Material transitions remain human-owned.

OpenCV evidence	Permitted action	Reason
No card over threshold	complete_no_material_change	No visual transition was extracted.
Card has no region	rerun_region_extraction_then_review	Do not summarize unlocalized evidence.
Localized card	seal_evidence_for_agent_summary	A later summary may cite the card.
Score ≥ 0.25	request_human_approval	A material transition requires a person.

FAIL-CLOSED CHECK

The explanation module rejects unknown frame citations and recommendations that do not exactly match the bounded plan.



SEEDED ENGINEERING SMOKE TEST

Small test. Explicit limits.

Four deterministic scenarios assert both event detection and the selected policy action.

Scenario	Expected event	Detected	Action
No change	No	No (TN)	complete
Localized change	Yes	Yes (TP)	seal
Full-frame change	Yes	Yes (TP)	human approval
Low-amplitude noise	No	No (TN)	complete

2 TP - 2 TN

0 false positives - 0 false negatives

4 / 4

expected actions matched

Public-sample checks

Frames	Events	Action
125	1	seal evidence
15	1	seal evidence
54	2	seal evidence

Limit: these are engineering gates and public-format checks, not a held-out real-world benchmark or production accuracy claim.



BOUNDED CLOUD DELIVERY

Private input. Encrypted evidence.

A live us-east-2 smoke run processed one right-cleared synthetic MP4, verified both receipts and structured logs, then removed every created AWS resource.

**RESPONSIBLE OPERATING BOUNDARY**

- One evidence card; 3,004 ms analysis; 3,791 ms billed.
- Encrypted 1,681-byte JSON; report and storage receipts verified.
- First cold initialization timed out; AWS retried and completed the event.
- No public bucket or API; full receipts are published in the README.
- Post-run checks: zero buckets, functions, ECR repos, active stacks, logs, or roles.



RESPONSIBLE AGENTIC VISION

See the change. Verify the action.

Evidence Locker is designed for inspection and review workflows where source frames, changed regions, and the reason for a next action must remain auditable.

Known limits

- Change detection is not object recognition, identity verification, intent inference, or a safety-critical anomaly detector.
- Thresholds are content-dependent; subtle changes can be missed and camera motion or illumination can trigger events.
- The 12-sequence foreground-change proxy (page 8) lacks natural negative videos. Production precision remains unproven.
- Hashes prove byte-level integrity, not the real-world meaning of a scene.
- Material transitions route to human approval; the agent cannot authorize irreversible actions.

REPRODUCE

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt

.venv/bin/python -m unittest -v

.venv/bin/python evaluate_synthetic.py

docker build --platform linux/arm64 -t evidence-locker .
```

SOURCE

github.com/tzh476/opencv-evidence-locker

NO AWARD OR PAYMENT CLAIMED · AWS SMOKE VERIFIED 2026-09-04 · RESOURCES REMOVED



FROZEN REAL-VIDEO PROBE

Real samples. Visible failures.

12 previously unused DAVIS validation sequences; three fixed frame pairs each. Protocol committed before predictions. Detector d09d968 and thresholds unchanged.

34 / 36

positive changes detected

2 missed

both in drift-chicane

0 negatives

real FPR not estimated

Reference: human foreground-mask occupancy XOR $\geq 0.5\%$ of pixels. Recall is 94.4% on this proxy; all 36 references are positive. This is not semantic incident accuracy or an official DAVIS score. Pairs within one sequence are correlated; hysteresis resets per pair.

Separate constructed controls: exact repeats 0/12 nuisance triggers; seeded +/-2 noise 0/12; +40 brightness 12/12. The detector is sensitive to light changes even when foreground geometry stays fixed.

Both misses remain published: drift-chicane frames 0 to 5 and 20 to 25. The online explorer exposes every prediction, independent annotation diagrams, source hashes, and a sealed JSON report. Two local runs produced identical artifacts.

Source: DAVIS 2017 (Pont-Tuset et al., arXiv:1704.00675; Perazzi et al., CVPR 2016). Annotation diagrams are adapted under CC BY 4.0 with attribution. Original camera pixels are not redistributed. Natural negative videos and operational labels remain future work.

tzh476.github.io/opencv-evidence-locker/evaluation/